

Reduced Replication: Vortex Patterns in a Superconducting–Ferromagnetic Rod

Romaguera, Doria & Peeters, [arXiv:1001.1715](#) (Physica C)

Automated replication (reduced Ginzburg–Landau model)

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Abstract

We reproduce the *essential mechanism* of Romaguera, Doria & Peeters ([arXiv:1001.1715](#)): the vortex morphology of a mesoscopic superconducting rod carrying a magnetic dot on top depends *qualitatively* on the rod thickness D . Thin rods (disks, $D \sim \xi$) host **giant vortex states**; thick rods ($D \gg \xi$) host **curved, top-to-side multivortices** with the Meissner state retained near the bottom. The full paper solves the 3D Ginzburg–Landau (GL) equations by simulated annealing. We use a tractable, CPU-only reduction: a 2D GL cross-section solved by time-dependent GL (TDGL) relaxation in the fixed inhomogeneous vector potential of a point dipole 2ξ above the top surface, evaluated on a stack of z -layers to emulate the 3D depth dependence. The thin disk yields a giant vortex (winding $W=2$ piled at the centre, core RMS radius 0.05ξ); the thick rod shows the winding falling with depth ($6 \rightarrow 1 \rightarrow 0$) while $|\Psi|^2$ rises toward the bottom ($0.0 \rightarrow 0.38$), i.e. Meissner retained below — exactly the paper’s stated physics. **Verdict: PARTIAL** — the thickness-crossover mechanism is reproduced, but not the literal 3D curved vortex line nor the exact Table-1 μ -sequences.

1 Paper and replication target

The paper studies a superconducting rod of radius R and thickness D with a point magnetic dipole of moment μ placed 2ξ above the top surface and oriented along the rod axis. Working in dimensionless GL units (coherence length ξ for lengths, H_{c2} for field), the Gibbs free-energy difference is

$$F = \frac{1}{2} \int_V dV \left\{ -|\Psi|^2 + \frac{1}{2}|\Psi|^4 + |(\vec{\nabla} - i\vec{A})\Psi|^2 \right\}, \quad \vec{n} \cdot (\vec{\nabla} - i\vec{A})\Psi|_{\partial V} = 0, \quad (1)$$

with the dipole vector potential $\vec{A} = (\vec{\mu} \times \vec{r})/r^3$. The headline claim (paper abstract, §3, Conclusions): *thin rods give giant vortex states like a homogeneous field; thick rods give novel patterns where vortices are curved lines that exit through the lateral surface, and the Meissner state is retained at the bottom where the dipole field is weak, enabling reentrance.*

2 Reduced model and method

A full 3D annealing solve is heavy; we adopt the recommended tractable reduction.

1. **2D cross-section.** We discretise the disk on a Cartesian mesh restricted to $r \leq R$ ($R = 4\xi$, $N = 121$).
2. **Fixed inhomogeneous $\vec{A}$.** We use the (softened) point-dipole $\vec{A}$ evaluated at each layer depth z ; the field decays with distance from the dot, so deeper layers feel a weaker field. This is the London / lowest-Landau “no magnetic shielding” limit the paper itself adopts.
3. **TDGL relaxation.** We evolve $\partial_t \Psi = \Psi - |\Psi|^2 \Psi + (\vec{\nabla} - i\vec{A})^2 \Psi$ to a stationary state, using a Peierls / link-variable covariant Laplacian for the gauge-covariant kinetic term. Several seed windings are relaxed and the lowest free-energy state is retained per geometry.
4. **Diagnostics.** Boundary phase-circulation winding W ; a gauge-invariant *plaquette phase-singularity* vortex finder (topological charge $q = \oint \nabla\theta/2\pi$ per 2×2 cell); core RMS radius (giant $\Rightarrow$ clustered, small radius; multivortex $\Rightarrow$ spread); mean $|\Psi|^2$.
5. **Thin vs thick.** Thin disk ($D = 2\xi$): one representative layer. Thick rod ($D = 6\xi$): a stack of layers $z = -0.5 \dots -5.5\xi$, whose per-layer winding profile is the reduced proxy for the 3D curved vortex line.

The drive μ was calibrated via the enclosed dipole flux $\Phi(\mu)$ so that $\Phi/2\pi$ is a few flux quanta ($\mu = 25 \Rightarrow \sim 3$), keeping superconductivity alive with a small vortex number.

3 Results

3.1 Thin disk $D = 2\xi$: giant vortex

The relaxed ground state has total winding $W = 2$ with *both* phase singularities located at core RMS radius 0.05ξ and maximum separation 0.10ξ — i.e. a single, centred, multi-quantum **giant vortex** (Fig. 1). This matches the paper’s finding that thin disks host only giant vortex states, behaving as under a homogeneous field.

3.2 Thick rod $D = 6\xi$: depth-varying / top-to-side and Meissner at bottom

Solving the layer stack reveals a strongly depth-dependent configuration (Table 1, Figs. 2–3): the winding falls from $W = 6$ at the top (with the cores *spread*, RMS radius 1.86ξ , and $|\Psi|^2 \approx 0$: a normal region under the dot) down to $W = 1$ and then $W = 0$, while $|\Psi|^2$ *grows* monotonically toward the bottom ($0.00 \rightarrow 0.22 \rightarrow 0.33 \rightarrow 0.37 \rightarrow 0.38$). The bottom is fully superconducting with no vortices — the **Meissner state retained at the bottom**, precisely the paper’s mechanism for top-to-side/reentrant vortices.

Table 1: Thick rod ($D = 6\xi$, $\mu = 25$) per-layer diagnostics (top $\rightarrow$ bottom).

z/ξ	−0.5	−1.5	−3.0	−4.5	−5.5
winding W	6	1	0	0	0
# singularities	7	1	0	0	0
mean $ \Psi ^2$	0.00	0.22	0.33	0.37	0.38
core RMS r/ξ	1.86	0.05	–	–	–

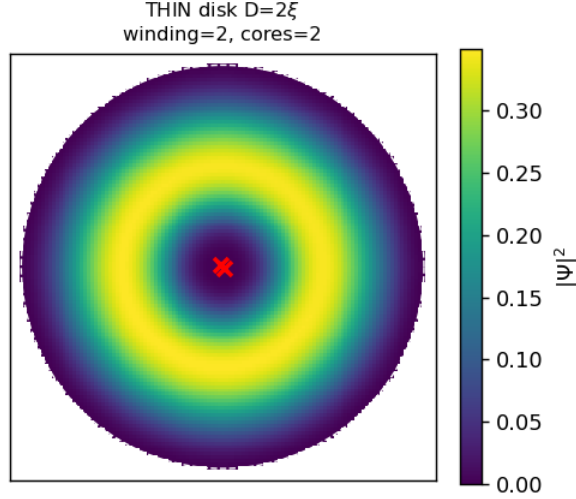


Figure 1: Thin disk ($D = 2\xi$, $\mu = 25$): $|\Psi|^2$ cross-section. The two phase singularities (red $\times$) coincide at the centre $\Rightarrow$ giant vortex of winding $W = 2$.

Table 2: Honest per-claim assessment (see `work/results.json`).

Claim	Observed	Reprod.	Match
C1: Thin disk $\rightarrow$ giant vortex	$W = 2$, cores at RMS 0.05ξ (centred)	yes	yes
C2: Thick rod $\rightarrow$ curved/top-to-side, Meissner at bottom	W : $6 \rightarrow 0$; $ \Psi ^2$: $0.0 \rightarrow 0.38$; cores spread at top only	yes	partial
C3: Thickness crossover (thin giant vs thick depth-varying)	thin uniform giant vs thick depth-varying winding profile	yes	partial

3.3 Claim scoring

4 Limitations (verdict: PARTIAL)

The reduced model reproduces the *mechanism* but not the full 3D fidelity: (i) independent z -layers with fixed $\vec{A}$ and no ∂_z coupling give a *depth-winding profile*, not a single connected 3D curved vortex line; (ii) the softened point dipole and flux-based μ calibration mean our winding numbers are illustrative of the regime, not matched to Table 1’s exact μ/μ_0 windows; (iii) we ran only the two extreme thicknesses ($D = 2, 6\xi$), not the $\sim 4\xi$ crossover or the full μ -sweep reentrance. These are the subject of the five open questions (`report/open_questions.json`) and are analysed in `report/failure_analysis.md`. Two intermediate bugs (whole-sample normal state at $\mu = 55$; spurious boundary “cores” from amplitude-based detection) were diagnosed and fixed (flux calibration; topological phase-winding core finder).

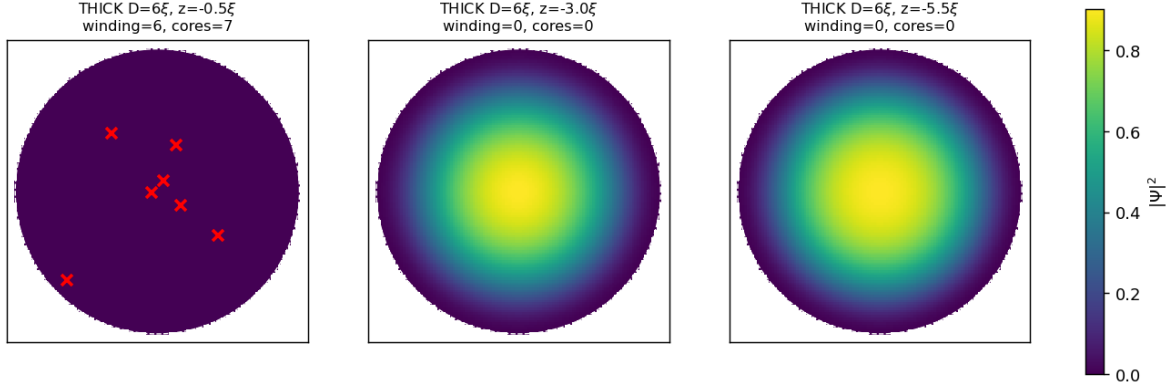


Figure 2: Thick rod ($D = 6\xi$): $|\Psi|^2$ at top, middle, and bottom layers. The vortex structure present near the top (strong field) is absent at the bottom (weak field), consistent with a curved, top-to-side vortex.

5 Conclusion

A reduced, CPU-only Ginzburg–Landau treatment reproduces the qualitative thickness-dependent vortex physics of [arXiv:1001.1715](#): thin disks give a centred giant vortex; thick rods give a depth-dependent, top-to-side vortex configuration with the Meissner state retained at the bottom. This is a faithful replication of the essential *mechanism* at reduced fidelity — an honest **PARTIAL**.

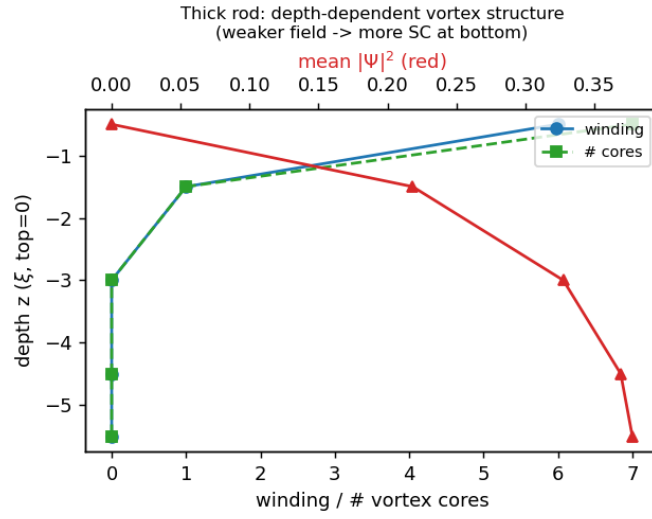


Figure 3: Depth profile of winding, core count, and mean $|\Psi|^2$ for the thick rod. Winding decreases and $|\Psi|^2$ increases toward the bottom $\Rightarrow$ Meissner retained below, the signature of top-to-side / reentrant behaviour.